

GENERALISED LAGUERRE SERIES FORMS OF WISHART DISTRIBUTIONS

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1. Introduction

This paper presents a technique of Laplace transform inversion in terms of Laguerre polynomials and applies this technique to obtain expansions of noncentral Wishart densities.

The usual meanings of some symbols are assumed and are given by Constantine (1966), who defines the generalised Laguerre polynomials by the expression

$$(1.1) \quad L_K^\gamma(S) = (\gamma + \frac{1}{2}(p+1))_K C_K(I_p) \sum_{j=0}^K \sum_J a_{K,J} \frac{(-1)^j C_J(S)}{(\gamma + \frac{1}{2}(p+1))_J C_J(I_p)},$$

where S is a $p \times p$ matrix.

The orthogonality property of Laguerre polynomials can be stated as

$$(1.2) \quad \int_{S \geq 0} L_K^\gamma(S) L_V^\gamma(S) g(S) dS = \begin{cases} 0 & \text{if } K \neq V \\ v! C_V(I_p) \Gamma_p(\gamma + \frac{1}{2}(p+1), V) & \text{if } K = V \end{cases}$$

where $g(S) = e^{\text{tr}(-S)} |S|^\gamma$

The polynomials in the elements of the matrix $S: p \times p$ reduce to the classical polynomials when $p=1$. Several results for the classical polynomials generalise to the case of matrix variables.

The following integrals are used in this paper.

Lemma 1.1: (James (1964))

$$F_q(a_1, \dots, a_r; b_1, \dots, b_q; S, Z) = \int_{O(p)} F_q(a_1, \dots, a_r; b_1, \dots, b_q; SHZH') dH$$

where the integral, taken over the orthogonal group $O(p)$,

$$\int_{O(p)} dH = 1, HH' = I_p$$

Lemma 1.2: (Constantine (1963))

$$\int_{S > 0} e^{\text{tr}(-RS)} |S|^{t-\frac{1}{2}(p+1)} C_K(SZ) dS = \Gamma_p(t) (t)_K |R|^{-t} C_K(ZR^{-1})$$

where R is a symmetric, positive definite matrix and Z is an arbitrary symmetric matrix.

Lemma 1.3: (Constantine (1966))

$$\int_{S > 0} e^{\text{tr}(-SR)} |S|^{\gamma} L_K^{\gamma}(S) dS = \Gamma_p(\gamma + \frac{1}{2}(p+1), K) |R|^{-\gamma - \frac{1}{2}(p+1)} C_K(I - R^{-1})$$

Lemma 1.4: (De Waal (1968))

$$\int_{S > 0} e^{\text{tr}(-S)} |S|^{t-\frac{1}{2}(p+1)} C_K(SZ) C_J(S) dS = \Gamma_p(t) \frac{C_K(Z)}{C_K(I_p)} \sum_{\delta} g_{K,J}^{\delta}(t)_{\delta} C_{\delta}(I_p)$$

Lemma 1.5:

$$\int_{S > 0} e^{\text{tr}(-S)} |S|^{t-\frac{1}{2}(p+1)} C_K(SZ) C_J(SW) dS = \Gamma_p(t) \sum_{\nu} P_{K,J}^{\nu}(Z, W) (t)_{\nu} C_{\nu}(I_p)$$

Proof:

Using the transformation $S \rightarrow H'SH$ and integrating over the orthogonal group $O(p)$, the integral becomes

$$\int_{S > 0} \int_{O(p)} e^{\text{tr}(-S)} |S|^{t-\frac{1}{2}(p+1)} C_K(SHZH') C_J(SHWH') dH dS$$

From Khatri (1971) it follows that

$$\int_{O(p)} C_K(SHZH') C_J(SHWH') dH = \sum_{\nu} C_{\nu}(S) P_{K,J}^{\nu}(Z, W)$$

where the summation is over all partitions ν of $k+j$ and $P_{K,J}^{\nu}(Z, W)$ depends only on Z and W .

Integration over S and the use of Lemma 1.2 then yield the result.

In section 2 we define (i) the moment generating function of a certain multivariate distribution and (ii) the moment generating function of a two-matrix multivariate distribution and find generalised Laguerre series for these distributions. In section 3 a series expansion of the multivariate Wishart distribution is written in terms of generalised Laguerre polynomials and section 4 defines the joint distribution of two symmetric matrices whose marginal distributions are Wishart. The results obtained appear important

in giving a simple method of approximating the expressions for large samples (Steyn and Roux (1972)) and to derive the distributions of some statistics. This is illustrated in section 5.

2. Generalised Laguerre series' for multivariate distributions

(i) We define the moment generating function of a certain multivariate distribution as

$$\begin{aligned} M^*(T) &= E(\exp \{ \text{tr} TX \}) \\ (2.1) \quad &= |I - 2\Sigma T|^{-1/2n} \int_0(p) r^F_q(a_1, \dots, a_r; b_1, \dots, b_q; \theta H' 2\Sigma T(I - 2\Sigma T)^{-1} H) dH \end{aligned}$$

where $\theta: p \times p$ is a positive definite matrix. If $\theta = 0$, it follows that $M^*(T) = M(T) = |I - 2\Sigma T|^{-1/2n}$ which is the moment generating function of a central Wishart distribution (Anderson (1958)) with covariance matrix Σ .

By using lemma (1.1) and the fact that (James (1964))

$$r^F_q(a_1, \dots, a_r; b_1, \dots, b_q; Z, W) = \sum_{k=0}^{\infty} \sum_K \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{C_K(Z) C_K(W)}{C_K(I_p) k!}$$

it follows from (2.1) that

$$(2.2) \quad M^*(T) = \sum_{k=0}^{\infty} \sum_K \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{C_K(-\theta) C_K(-2\Sigma T(I - 2\Sigma T)^{-1})}{C_K(I_p) k!} |I - 2\Sigma T|^{-1/2n}.$$

Using the Laplace inverse, namely

$$f^*(X) = \frac{2^{1/2p(p-1)}}{(2\pi i)^{1/2p(p+1)}} \int_{R(T) > 0} \exp \{ -\text{tr} TX \} M^*(T) dT$$

it follows from Constantine (1966) and (2.2) that

$$f^*(X) = f(X) \sum_{k=0}^{\infty} \sum_K \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{1}{(1/2n)_K} \frac{C_K(-\theta)}{C_K(I_p) k!} L_K^{1/2(n-p-1)} (1/2\Sigma^{-1} X)$$

where

$$f(X) = \frac{1}{|2\Sigma|^{1/2n} \Gamma_p(1/2n)} e^{-1/2\text{tr}\Sigma^{-1}X} |X|^{1/2(n-p-1)}, \quad X: p \times p > 0$$

is a Wishart probability density function (Anderson (1958)).

(ii) We define

$$(2.3) \quad M(T_1, T_2) = |I - 2\Sigma_1 T_1|^{-1/2n_1} |I - 2\Sigma_2 T_2|^{-1/2n_2} \\ \cdot {}_1F_q(a_1, \dots, a_r; b_1, \dots, b_q; \theta, 2\Sigma_1 T_1(I - 2\Sigma_1 T_1)^{-1}, 2\Sigma_2 T_2(I - 2\Sigma_2 T_2)^{-1})$$

as a possible moment generating function for a two-matrix multivariate distribution in which each of the random matrices has a Wishart probability density function

$$f_{n_i}(X_i) = \frac{1}{|2\Sigma_i|^{1/2n_i} \Gamma_p(1/2n_i)} e^{\text{tr} - 1/2 \Sigma_i^{-1} X_i} |X_i|^{1/2(n_i - p - 1)} \quad \text{for } X_i: p \times p > 0, i = 1, 2.$$

From (2.3) and the fact that (James (1964))

$${}_1F_q(a_1, \dots, a_r; b_1, \dots, b_q; Z, W, Q) = \sum_{k=0}^{\infty} \sum_K \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{C_K(Z) C_K(W) C_K(Q)}{(C_K(I_p))^{2k}}$$

it follows that

$$(2.4) \quad M(T_1, T_2) = \sum_{k=0}^{\infty} \sum_K \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{C_K(\theta)}{(C_K(I_p))^{2k}} |I - 2\Sigma_1 T_1|^{-1/2n_1} \\ \cdot C_K(2\Sigma_1 T_1(I - 2\Sigma_1 T_1)^{-1}) |I - 2\Sigma_2 T_2|^{-1/2n_2} C_K(2\Sigma_2 T_2(I - 2\Sigma_2 T_2)^{-1}).$$

Since $\Gamma_p(n, K) |I - T|^{-n} C_K(-T(I - T)^{-1})$ is the moment generating function of

$$e^{\text{tr} - X|X|^{n-1/2(p+1)}} L_K^{n-1/2(p+1)}(X) \quad (\text{Lemma 1.3})$$

it follows that (2.4) is the moment generating function of

$$f(X_1, X_2) = f_{n_1}(X_1) f_{n_2}(X_2) \left[\sum_{k=0}^{\infty} \sum_K \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{C_K(\theta)}{(C_K(I_p))^{2k}} \right. \\ \left. \cdot \frac{1}{(\frac{1}{2}n_1)_K (\frac{1}{2}n_2)_K} L_K^{1/2(n_1 - p - 1)} (\frac{1}{2}\Sigma_1^{-1} X_1) L_K^{1/2(n_2 - p - 1)} (\frac{1}{2}\Sigma_2^{-1} X_2) \right].$$

3 Laguerre series of the noncentral Wishart distribution

Let $A: p \times p$, be a noncentral Wishart matrix with parameter n , covariance matrix Σ

and noncentrality parameter $\Omega = \frac{1}{2}\mu\mu'\Sigma^{-1}$. Then the probability density function, $f^*(A)$, is given by (Constantine (1963))

$$f^*(A) = \frac{|A|^{\frac{1}{2}(n-p-1)} e^{-\frac{1}{2}\text{tr}\Sigma^{-1}A} e^{\text{tr}(-\Omega)}}{|2\Sigma|^{\frac{1}{2}n} \Gamma_p(\frac{1}{2}n)} {}_0F_1(\frac{1}{2}n; \frac{1}{2}\Sigma^{-1}\Omega A).$$

Consider a formal expansion (Steyn and Roux (1972) and section 2)

$$(3.1) \quad f^*(A) = f(A) \sum_{k=0}^{\infty} \sum_K \alpha_K L_K^{\frac{1}{2}(n-p-1)} (\frac{1}{2}\Sigma^{-1}A)$$

if the series converges, where $f(A)$ is the probability density function of the central Wishart distribution (Anderson (1958)) with parameter n and the α_K are constants.

Multiply both sides of (3.1) by $L_K^{\frac{1}{2}(n-p-1)} (\frac{1}{2}\Sigma^{-1}A)$ and integrate over A , when

$$\begin{aligned} & \int_{A>0} f^*(A) L_K^{\frac{1}{2}(n-p-1)} (\frac{1}{2}\Sigma^{-1}A) dA \\ &= \int_{A>0} f(A) L_K^{\frac{1}{2}(n-p-1)} (\frac{1}{2}\Sigma^{-1}A) \sum_{k=0}^{\infty} \sum_K \alpha_K L_K^{\frac{1}{2}(n-p-1)} (\frac{1}{2}\Sigma^{-1}A) dA \end{aligned}$$

that is

$$E(L_K^{\frac{1}{2}(n-p-1)} (\frac{1}{2}\Sigma^{-1}A)) = \alpha_K C_K(I_p) (\frac{1}{2}n)_K k!$$

by virtue of the orthogonality property (1.2).

Thus

$$(3.2) \quad \alpha_K = \frac{E(L_K^{\frac{1}{2}(n-p-1)} (\frac{1}{2}\Sigma^{-1}A))}{C_K(I_p) (\frac{1}{2}n)_K k!}, \text{ for } K = 0; 1; 2; 1^2; 3; \dots$$

From equation (1.1) it follows that

$$\begin{aligned} (3.3) \quad E(L_K^{\frac{1}{2}(n-p-1)} (\frac{1}{2}\Sigma^{-1}A)) &= (\frac{1}{2}n)_K C_K(I_p) \sum_{j=0}^k \sum_J a_{K,J} \frac{(-1)^j}{(\frac{1}{2}n)_J C_J(I_p)} \\ &\quad \cdot E(C_J(\frac{1}{2}\Sigma^{-1}A)) \end{aligned}$$

and

$$E(C_J(\frac{1}{2}\Sigma^{-1}A)) = \int_{A>0} f^*(A) C_J(\frac{1}{2}\Sigma^{-1}A) dA.$$

Using the transformation $\frac{1}{2}\Sigma^{-1/2}A\Sigma^{-1/2} \rightarrow A$ where the jacobian is $|2\Sigma|^{1/2(p+1)}$ it follows that

$$(3.4) \quad E(C_J(\frac{1}{2}\Sigma^{-1}A)) = \frac{1}{\Gamma_p(\frac{1}{2}n)} e^{\text{tr}(-\Omega)} \sum_{v=0}^{\infty} \sum_V \frac{1}{(\frac{1}{2}n)_V} \frac{1}{v!} \int_{A>0} |A|^{1/2(n-p-1)} \\ \cdot e^{-\text{tr} A} C_{V(\eta A)} C_J(A) dA$$

where $\eta = \frac{1}{2}\Sigma^{-1/2}\mu\mu'\Sigma^{-1/2}$.

Using lemma 1.4 it follows that

$$E(C_J(\frac{1}{2}\Sigma^{-1}A)) = e^{\text{tr}(-\Omega)} \sum_{v=0}^{\infty} \sum_V \frac{1}{(\frac{1}{2}n)_V} \frac{1}{v!} \frac{C_V(\Omega)}{C_V(I_p)} \sum_{\delta} g_{V,J}^{\delta} (\frac{1}{2}n)_{\delta} C_{\delta}(I_p).$$

It now follows from (3.2), (3.3) and (3.4) that

$$\alpha_K = \frac{1}{k!} \sum_{j=0}^k \sum_J a_{K,J} \frac{(-1)^j}{(\frac{1}{2}n)_J C_J(I_p)} e^{\text{tr}(-\Omega)} \sum_{v=0}^{\infty} \sum_V \frac{1}{(\frac{1}{2}n)_V} \frac{1}{v!} \frac{C_V(\Omega)}{C_V(I_p)} \\ \sum_{\delta} g_{V,J}^{\delta} (\frac{1}{2}n)_{\delta} C_{\delta}(I_p).$$

4. Series' in generalised Laguerre polynomials

Let A_i : $p \times p$ be Wishart matrices with parameters n_i and covariance matrices Σ_i , $i=1,2$. The density function of the joint distribution, $f(A_1, A_2)$, can be formally expanded as a sum of generalised Laguerre polynomials, namely (see section 2)

$$(4.1) \quad f(A_1, A_2) = f_{n_1}(A_1) f_{n_2}(A_2) \left\{ \sum_{k=0}^{\infty} \sum_K \beta_K L_K^{\frac{1}{2}(n_1-p-1)} (\frac{1}{2}\Sigma_1^{-1}A_1) \right. \\ \left. \cdot L_K^{\frac{1}{2}(n_2-p-1)} (\frac{1}{2}\Sigma_2^{-1}A_2) \right\}$$

where $f_{n_i}(A_i)$ is the probability density function of the central Wishart matrix A_i with parameter n_i , and the β_K are constants. We again impose the condition that the series must converge.

Definition 4.1

Let $A_1: p \times p$ and $A_2: p \times p$ be two Wishart matrices with parameters n_1 and n_2 respectively. (A_1, A_2) will be said to be distributed in Wishart correlation if

$$E(L_K^{1/2(n_1-p-1)}(A_1)L_K^{1/2(n_2-p-1)}(A_2)) = \rho_K,$$

$\{\rho_K\}$ being a sequence of constants. (Griffiths (1969)).

Multiplying both sides of (4.1) by $L_K^{1/2(n_1-p-1)}(1/2\Sigma_1^{-1}A_1)$ $L_K^{1/2(n_2-p-1)}(1/2\Sigma_2^{-1}A_2)$ and integrating over A_1 and A_2 , we obtain

$$\beta_K = \frac{\rho_K'}{(C_K(I_p))^2 (1/2n_1)_K (1/2n_2)_K (k!)^2}$$

To show that $\beta_K^2 \leq 1$ we have, using the Cauchy-Schwartz inequality, that

$$\begin{aligned} & \{E(L_K^{1/2(n_1-p-1)}(1/2\Sigma_1^{-1}A_1)L_K^{1/2(n_2-p-1)}(1/2\Sigma_2^{-1}A_2))\}^2 \\ & \leq E(L_K^{1/2(n_1-p-1)}(1/2\Sigma_1^{-1}A_1))^2 E(L_K^{1/2(n_2-p-1)}(1/2\Sigma_2^{-1}A_2))^2 \end{aligned}$$

that is

$$\rho_K'^2 \leq (k! C_K(I_p))^2 (1/2n_1)_K (1/2n_2)_K$$

so that

$$\beta_K^2 \leq 1$$

The moment generating function of $f(A_1, A_2)$ is given by

$$\begin{aligned} M(T_1, T_2) &= E(e^{\text{tr}(T_1 A_1 + T_2 A_2)}) = \int_{A_1 > 0} \int_{A_2 > 0} e^{\text{tr}(T_1 A_1 + T_2 A_2)} f_{n_1}(A_1) \\ & \quad \cdot f_{n_2}(A_2) \sum_{k=0}^{\infty} \sum_K \beta_K L_K^{1/2(n_1-p-1)}(1/2\Sigma_1^{-1}A_1) \\ & \quad \cdot L_K^{1/2(n_2-p-1)}(1/2\Sigma_2^{-1}A_2) dA_1 dA_2 \end{aligned}$$

and by using Lemma 1.3 it follows that

$$\begin{aligned} M(T_1, T_2) &= |I - 2\Sigma_1 T_1|^{-1/2n_1} |I - 2\Sigma_2 T_2|^{-1/2n_2} \sum_{k=0}^{\infty} \sum_K \beta_K (1/2n_1)_K (1/2n_2)_K \\ & \quad \cdot C_K(2\Sigma_1 T_1 (I - 2\Sigma_1 T_1)^{-1}) C_K(2\Sigma_2 T_2 (I - 2\Sigma_2 T_2)^{-1}). \end{aligned}$$

5. Examples of uses of the generalised Laguerre series

(i) A_1 is distributed according to a Wishart distribution with parameter n_1 and A_2 is distributed independently according to a noncentral Wishart with parameter n_2 and noncentrality parameter $\Omega = \frac{1}{2}\mu\mu'\Sigma^{-1}$. Let

$$L = G^{-1/2}A_1G^{-1/2}, G = A_1 + A_2 \text{ and } J(A_1 \rightarrow L) = |G|^{1/2(p+1)}$$

The joint distribution of A_1 and A_2 is given by

$$(5.1) \quad f(A_1, A_2) = \frac{|A_1|^{1/2(n_1-p-1)} |A_2|^{1/2(n_2-p-1)} e^{-\frac{1}{2}\text{tr}\Sigma^{-1}(A_1+A_2)}}{|2\Sigma|^{1/2(n_1+n_2)} \Gamma_p(1/2n_1) \Gamma_p(1/2n_2)} \\ \cdot \sum_{k=0}^{\infty} \sum_K \alpha_K L_K^{1/2(n_2-p-1)} (\frac{1}{2}\Sigma^{-1}A_2)$$

Thus from (5.1) and (1.1) the joint distribution of L and G follows as

$$(5.2) \quad f^*(L, G) = \frac{|L|^{1/2(n_1-p-1)} |I-L|^{1/2(n_2-p-1)}}{|2\Sigma|^{1/2(n_1+n_2)} \Gamma_p(1/2n_1) \Gamma_p(1/2n_2)} |G|^{1/2(n_1+n_2-p-1)} e^{-\frac{1}{2}\text{tr}\Sigma^{-1}G} \\ \cdot \sum_{k=0}^{\infty} \sum_K \alpha_K (\frac{1}{2}n_2)_K C_K(I_p) \sum_{j=0}^k \sum_J a_{K,J} \frac{(-1)^j C_J(\frac{1}{2}\Sigma^{-1}G^{1/2}(I-L)G^{1/2})}{(\frac{1}{2}n_2)_J C_J(I_p)}.$$

The density (5.2) may also be written as (Kabe (1972))

$$f(L, G) = \frac{|L|^{1/2(n_1-p-1)} |I-L|^{1/2(n_2-p-1)}}{2^{1/2p(n_1+n_2)} |\Sigma|^{1/2(p+1)} \Gamma_p(1/2n_1) \Gamma_p(1/2n_2)} |QG^{1/2}\Sigma^{-1}G^{1/2}Q'|^{1/2(n_1+n_2-p-1)} \\ \cdot e^{-\frac{1}{2}\text{tr}QG^{1/2}\Sigma^{-1}G^{1/2}Q'} \sum_{k=0}^{\infty} \sum_K \alpha_K (\frac{1}{2}n_2)_K C_K(I_p) \\ \cdot \sum_{j=0}^k \sum_J a_{K,J} \frac{(-1)^j C_J(\frac{1}{2}\Delta^2 Q(I-L)Q^{-1})}{(\frac{1}{2}n_2)_J C_J(I_p)}$$

where Δ^2 is the diagonal matrix of the roots of $|\Sigma^{-1}G - \lambda(I-L)^{-1}| = 0$ and Q is an arbitrary orthogonal matrix.

Denoting $QG^{1/2}\Sigma G^{1/2}Q'$ by $G^{1/2}\Sigma G^{1/2}$ itself and setting $B = Q^{-1}H$ we find that

$$f(L, G) = \frac{|L|^{1/2(n_1-p-1)} |I-L|^{1/2(n_2-p-1)}}{|2\Sigma|^{1/2(n_1+n_2)} \Gamma_p(1/2 n_1) \Gamma_p(1/2 n_2)} |G|^{1/2(n_1+n_2-p-1)} e^{-1/2 \text{tr} \Sigma^{-1} G} \\ \cdot \sum_{k=0}^{\infty} \sum_K \alpha_K(1/2 n_2)_K C_K(I_p) \sum_{j=0}^k \sum_J a_{K,J} \frac{(-1)^j C_J(1/2 \Sigma^{-1} G B(I-L)B')}{(1/2 n_2)_J C_J(I_p)}$$

where H is an arbitrary orthogonal matrix.

Integrating w.r.t. G , and using Lemma 1.2 it follows that

$$f^*(L) = f(L) \sum_{k=0}^{\infty} \sum_K \sum_{j=0}^k \sum_J \alpha_{K,J}^* C_J(I-L)$$

where $L = BLB'$,

$$f(L) = \frac{\Gamma_p(1/2(n_1+n_2))}{\Gamma_p(1/2 n_1) \Gamma_p(1/2 n_2)} |L|^{1/2(n_1-p-1)} |I-L|^{1/2(n_2-p-1)}$$

$$\text{and } \alpha_{K,J}^* = \alpha_K(1/2 n_2)_K C_K(I_p) a_{K,J} \frac{(-1)^j}{(1/2 n_2)_J C_J(I_p)} (1/2(n_1+n_2))_J$$

(ii) Consider the joint distribution of A_1 and A_2 as defined in (4.1), then the joint distribution of L and G is given by

$$f^{**}(L, G) = \frac{|G|^{1/2(n_1-p-1)} |L|^{1/2(n_1-p-1)} e^{-1/2 \text{tr} \Sigma_1^{-1} G^{1/2} L G^{1/2}}}{|2\Sigma_1|^{1/2 n_1} \Gamma_p(1/2 n_1)} \\ \cdot \frac{|G|^{1/2(n_2-p-1)} |I-L|^{1/2(n_2-p-1)} e^{-1/2 \text{tr} \Sigma_2^{-1} G^{1/2} (I-L) G^{1/2}}}{|2\Sigma_2|^{1/2 n_2} \Gamma_p(1/2 n_2)} \\ \cdot |G|^{1/2(p+1)} \sum_{k=0}^{\infty} \sum_K \beta_K L_K^{1/2(n_1-p-1)} (1/2 \Sigma_1^{-1} G^{1/2} L G^{1/2}) \\ \cdot L_K^{1/2(n_2-p-1)} (1/2 \Sigma_2^{-1} G^{1/2} (I-L) G^{1/2}).$$

Consider the special case where $\Sigma_1 = \Sigma_2 = \Sigma$, then by using (1.1) and Lemma 1.5 it follows that

$$f^{**}(L) = f(L) \sum_{k=0}^{\infty} \sum_K \sum_{j=0}^k \sum_J \sum_{v=0}^k \sum_V \sum_{\nu} \beta_{K,J,V,\nu}^* P_{J,V}^{\nu}(L, I-L)$$

where

$$\begin{aligned} \beta_{K,J,V,\nu}^* &= \beta_K^{(1/2n_1)} \beta_K^{(1/2n_2)} (C_K(I_p))^2 a_{K,J} a_{K,V} \frac{(-1)^{j+v}}{(1/2n_1)_J (1/2n_2)_V C_J(I_p) C_V(I_p)} \\ &\quad \cdot 2^{j+v} (1/2(n_1+n_2))_{\nu} C_{\nu}(I_p), \end{aligned}$$

$L = BLB'$ and B is an arbitrary orthogonal matrix.

(iii) Similarly, we can derive the distribution of $F = A_2^{-1/2} A_1 A_2^{-1/2}$ as in (i) and (ii) respectively.

$$f^*(F) = f(F) \sum_{k=0}^{\infty} \sum_K \sum_{j=0}^{\infty} \sum_J \gamma_{K,J}^* C_J((I+F)^{-1})$$

and

$$f^{**}(F) = f(F) \sum_{k=0}^{\infty} \sum_K \sum_{j=0}^k \sum_J \sum_{v=0}^k \sum_V \sum_{\delta} \gamma_{K,J,V,\delta}^{**} C_J(F) C_{\delta}((I+F)^{-1})$$

where $F = BFB'$,

$$f(F) = \frac{\Gamma_p(1/2(n_1+n_2))}{\Gamma_p(1/2n_1) \Gamma_p(1/2n_2)} |F|^{1/2(n_1-p-1)} |I+F|^{-1/2(n_1+n_2)},$$

$$\gamma_{K,J}^* = \alpha_K^{(1/2n_2)} \beta_K^{(1/2n_1)} a_{K,J}^{(1/2(n_1+n_2))_J} \frac{(-1)^j}{(1/2n_2)_J C_J(I_p)}$$

and

$$\begin{aligned} \gamma_{K,J,V,\delta}^{**} &= \beta_K^{(1/2n_1)} \beta_K^{(1/2n_2)} (C_K(I_p))^2 a_{K,J} a_{K,V} \\ &\quad \cdot \frac{(-1)^{j+k}}{(1/2n_1)_J (1/2n_2)_V C_J(I_p) C_V(I_p)} g_{J,V}^{\delta} (1/2(n_1+n_2))_{\delta} \end{aligned}$$

(iv) The distribution of $T = \text{tr} A_1^{-1} A_2$ will be defined by inverting the Laplace transform

$$g(t) = E [e^{-\text{tr } t A_1^{-1} A_2}]$$

of its density function. (Constantine (1966)). Multiplying (5.1) by $e^{-\text{tr } t A_1^{-1} A_2}$ and integrating over $A_2 > 0$ and $A_1 > 0$, it follows that

$$g(t) = \frac{\Gamma_p(\frac{1}{2}(n_1+n_2))}{\Gamma_p(\frac{1}{2}n_1)} \sum_{k=0}^{\infty} \sum_K \sum_{m=0}^{\infty} \sum_{\eta} \sum_{\delta} \alpha_K \frac{(-1)^m}{(\frac{1}{2}n_2)_K} (\frac{1}{2}n_2)_{\delta} g_{K,\eta}^{\delta} \frac{C_{\delta}(I_p)}{m!} \\ \cdot t^{-\frac{1}{2}pn_2-m} (\frac{1}{2}(n_1+n_2))_{\eta}.$$

$$\text{Since } \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{tT} t^{-\frac{1}{2}pn_2-m} dt = \frac{1}{\Gamma(\frac{1}{2}pn_2+m)} T^{\frac{1}{2}pn_2+m-1}$$

we have

$$f(T) = \frac{\Gamma_p(\frac{1}{2}(n_1+n_2))}{\Gamma_p(\frac{1}{2}n_1)} \sum_{k=0}^{\infty} \sum_K \sum_{m=0}^{\infty} \sum_{\eta} \sum_{\delta} \alpha_K \frac{(-1)^m}{(\frac{1}{2}n_2)_K} (\frac{1}{2}n_2)_{\delta} g_{K,\eta}^{\delta} \frac{C_{\delta}(I_p)}{m!} \\ \cdot \frac{(\frac{1}{2}(n_1+n_2))_{\eta} T^{\frac{1}{2}pn_2+m-1}}{\Gamma(\frac{1}{2}pn_2+m)}$$

(v) For large values of the noncentrality parameter, the series (3.1) converges slowly. The approximation of the noncentral Wishart by a central Wishart is therefore not satisfactory for large values of Ω . However, for constant values of Ω and large n , it gives a good approximation.

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